一. 填空

1. $\overline{Z}=(A+\overline{B}\cdot C\overline{D})(B+\overline{C})$

2. $R=60\Omega$ 欠阻尼

3. N 耗尽型

4. A

5. 6位 100/001

6. 8根 8根

7. (1) × 会出现Q=$\overline{Q}$=0，无意义

(2) √ (?我也不确定)

(3) ×

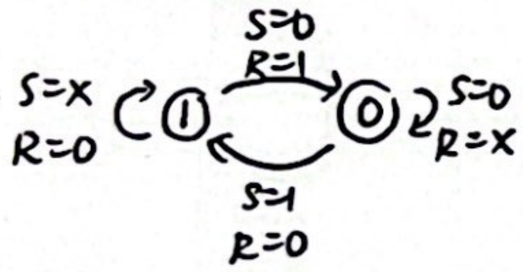
(4) √ (都是时序电路，只是ROM不随时间改变 → 里的信息)

二. 简答

1. C=0，CMOS，L=B

C=1，TTL，L=$\overline{B}$

2. $L=A\oplus B$

3. 

4. $I_B=\frac{6.7-0.7}{(1+\beta)R_e}\approx\frac{6V}{\beta R_e}=0.03mA$

假设放大 $I_C=\beta I_B=3mA$

$V_{CE}=V_{CC}-I_C(R_c+R_e)=-3V<0$

∴假设不成立，处于饱和.

$V_{CE}=V_{CES}=0.3V$

$I_C=\frac{V_{CC}-V_{CE}-U_e}{R_e}=\frac{12-0.3-6}{3k\Omega}=1.9mA$

三. ①直流 $U_{S(0)}=15V$，$I_{(0)}=0.1A$，$P_{W1(0)}=0W$，$P_{W2(0)}=1.5W$

②基波 $i_{S(1)}(t)=2\sqrt{2}\sin(1000t)A$.

$\dot{I}_{S(1)}=2\angle0°A$

$\dot{I}_{(1)}=\frac{\dot{I}_{S(1)}\cdot j\omega_1M}{R_2+j\omega_1(L_2+L)}=\frac{1}{\sqrt{10}}\angle-108.4°A$

$\dot{U}_{W1(1)}=\dot{I}_{S(1)}(j\omega_1L-j\frac{1}{\omega_1C})+j\omega_1M\dot{I}_{(1)}=\frac{25}{\sqrt{10}}\angle-18.4°V$

消耗感.

$P_{W1(1)}=15W$，$P_{W2(1)}=0W$.

③3次谐波 $\dot{U}_{S(3)}=30\angle0°V$.

$\dot{I}_{(3)}=\frac{\dot{U}_{S(3)}}{R_2+j\omega_3L_2+j\omega_3L}=\frac{1}{5\sqrt{2}}\angle-45°A$

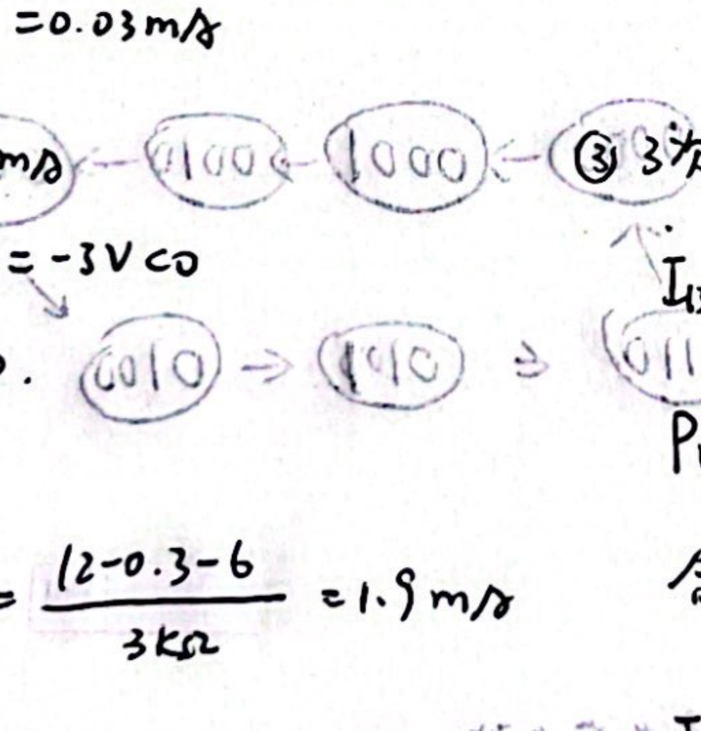
$P_{W1(3)}=0W$，$P_{W2(3)}=3W$.

合成：$i(t)=0.1+\frac{\sqrt{2}}{\sqrt{10}}\sin(1000t-108.4°)+\frac{1}{5}\sin(3000t$

$I=\sqrt{0.1^2+(\frac{1}{\sqrt{10}})^2+(\frac{1}{5\sqrt{2}})^2}=\frac{\sqrt{13}}{10}A=0.361A$

四.

Q_1 Q_0	P_1 P_2 P_3 P_4 P_5 P_6
0 0	0 0 0 0 1 1
0 1	1 1 1 1 0 0
1 0	1 1 1 1 1 0
1 1	1 1 1 1 1 1



卡诺图略 $P_1=P_2=P_3=P_4=Q_0+Q_1$

$P_5=Q_1+\overline{Q_0}$

$P_6=Q_1Q_0+\overline{Q_1}\overline{Q_0}$

功率表 $W_1=4.5W$，$W_2=4.5W$

五.

$Q_1Q_0^n$	$Q_1^{n+1}Q_0^{n+1}$
0 0	1 0
1 0	0 1
0 1	1 1
1 1	0 0

Q_1^{n+1}

Q_1^n \ Q_0^n	0	1
0	1	1
1	0	0

$Q_1^{n+1}=Q_1^n$

$\because Q_1^{n+1}=J\overline{Q_1^n}+\overline{K}Q_1^n$

$J_1=K_1=1$

Q_0

Q_1^n \ Q_0^n	0	1
0	0	1
1	1	0

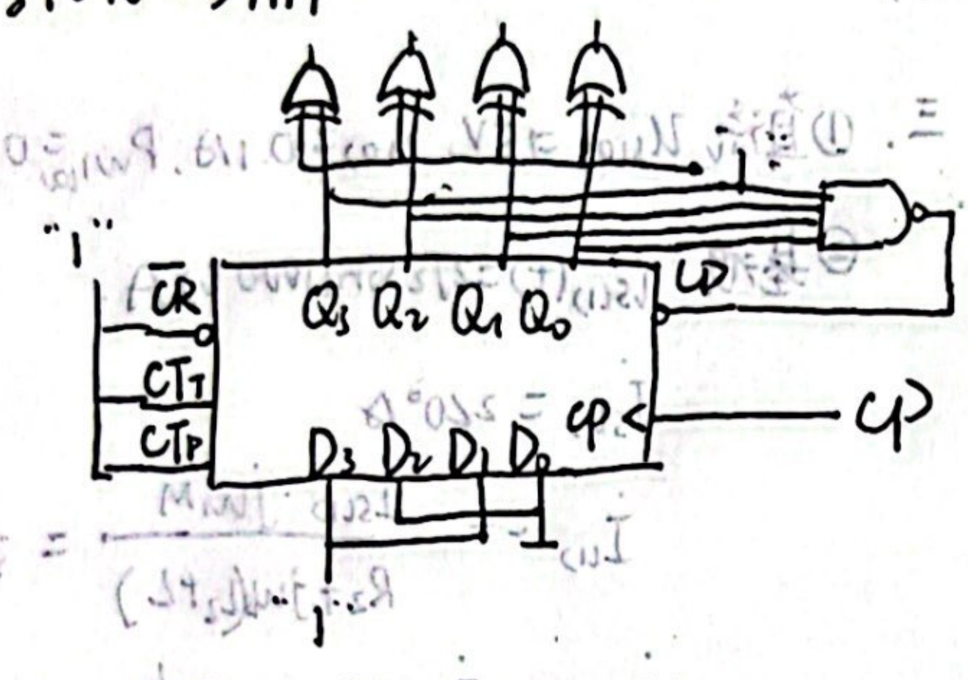
$Q_0^{n+1}=Q_1^n\overline{Q_0^n}+\overline{Q_1^n}Q_0^n$

$Q_0^{n+1}=J_0\overline{Q_0^n}+\overline{K}Q_0^n$

$J_0=Q_1^n,\ K=Q_1^n$ 补图连线略.

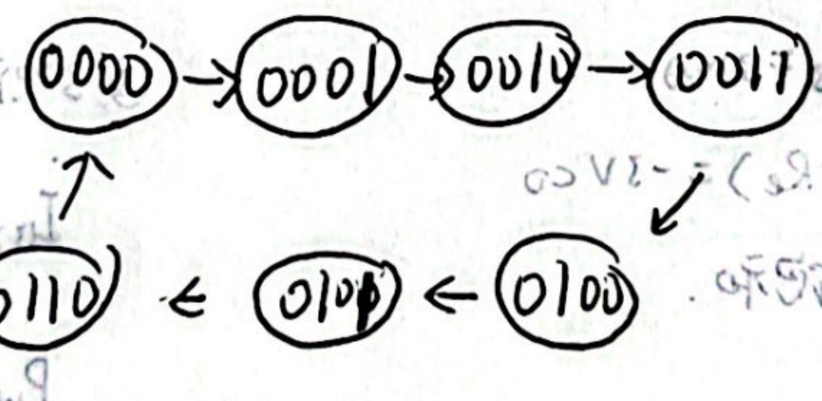
六、 1. 0101→0000

反馈1010→1111



2. 实现 $(I_3I_2I_1I_0+1)_{8421BCD}$ 进制用加法计数器功能.

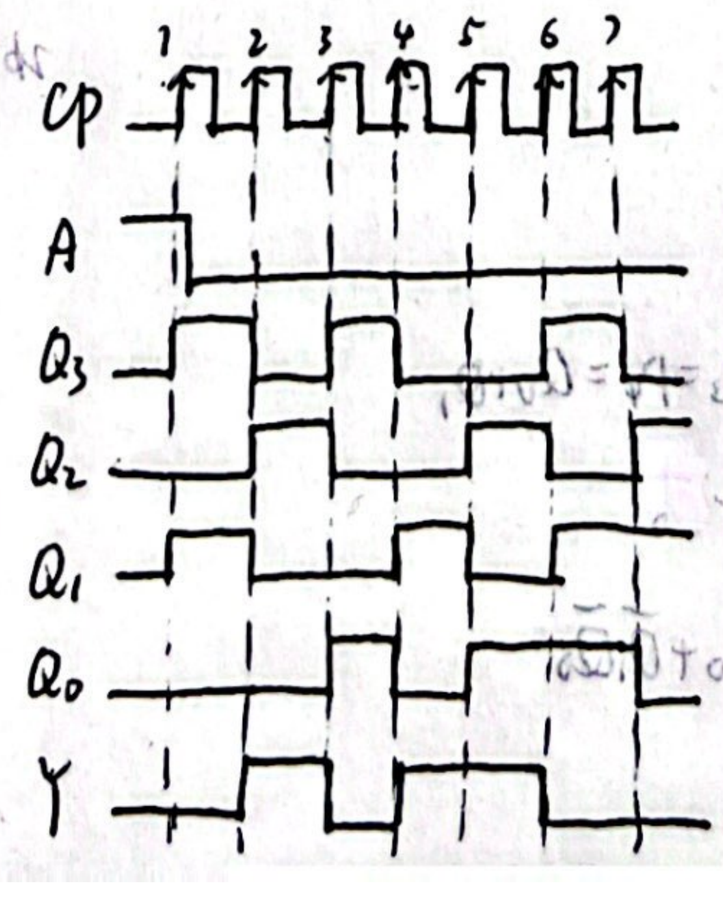
$A_3A_2A_1A_0$	$\overline{CR}$
0 1 1 0	0
0 0 0 0	1
0 1 0 1	1
0 1 1 0	1
0 0 0 0	1



七. (1) $Y=\overline{\overline{Y_5Y_2Y_1}}=Y_5+Y_2+Y_1$ 移位寄存器右移, $D_{SR}=Y=Y_5+Y_2+Y_1$

(2)

CP	A	$A_2A_1A_0$	E	Y
0		0 0 0	0	0
1↑	1	0 1 0	1	0
2↑	0	0 0 1	0	1
3↑	0	1 0 0	1	0
4↑	0	0 1 0	0	1
5↑	0	1 0 1	0	1
6↑	0	1 1 0	1	0
7↑	0	0 1 1	0	0



八. 可重角出发电路. $T_W = T_{W1} + t = T_{W1} + RC\ln 3 = 1+2\times 0.47\times \ln 3 = 2.03ms$.

